# Base paper questions

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PDF

this is my base paper .now  from it i will ask you questions .answer that .

Great! I've loaded your base paper completely and I'm ready.

You can now ask **any question** from the paper — concepts, formulas, explanations, diagrams, algorithms, comparison tables — *anything*.
I'll answer clearly and simply, exactly the way you need for your exam or understanding.

Just send your first question.

tell me summery of the introduction part

Here is a **simple and clear summary** of the *Introduction* section of your base paper:

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### Summary of the Introduction (Easy to Understand)

The paper talks about how tourism in China is growing fast, especially after COVID-19. More people now prefer **self-planned trips** like backpacking and road trips. But planning such trips is difficult because:

- There is **too much information online**,
- Tourists must consider many things (time, distance, transport, crowding, etc.),
- It becomes confusing for someone visiting a new place.

Because of this, we need a smart system that can quickly plan an **optimal travel route**.

Researchers have been using different optimization algorithms (like ant colony, genetic algorithm, particle swarm, simulated annealing) to solve route-planning problems.

Among these, the **Ant Colony Optimization (ACO)** algorithm is widely used because it can find good paths, just like real ants find food.

However, traditional ACO has problems:

- Slow convergence (takes many iterations)
- May get stuck in poor solutions
- Lower global search ability

So, this paper proposes a **new improved model**:

### GA-PSACO (Genetic Algorithm + Path-Smoothing Ant Colony Optimization)

The improved model:

- Combines **genetic algorithm (GA)** with ACO,
- Adds **smoothing factor** and **guiding factor** to speed up convergence,
- Makes the searching process faster and more accurate,
- Helps plan the **best travel route** that matches tourists' needs.

The main goal of the paper is to create a **smart tourism route planning system** that:

- Finds the best path quickly,
- Considers tourist preferences,
- Reduces planning time,
- Improves travel experience.

---

If you want, I can also summarize the next section (Related Works) or explain any formulas.

Currently, there is even a rise in the
special forces-style and military training style of traveling to several sites
in one day, putting forward higher requirements for personalized travel
itinerary planning (Song et al., 2022). When tourists plan their travel
itineraries independently, they need to obtain relevant contents from a

large amount of network information. The problem of tourist informa
tion overload caused by the network sharing mechanism and the various
constraints that need to be considered in the itinerary make tourists
spend a lot of energy in screening the appropriate tourist services. This is
certainly a complex dilemma for tourists visiting unfamiliar cities. If an
optimal travel route can be quickly planned for a journey with many
attractions and a short duration, it will not only enhance the consumer's
journey experience, but also be an unbeatable benefit for the attractions.  tell me it in short and precise

Here is a **short and precise** version:

People now try to visit many places in one day, which requires more personalized and efficient trip planning. But when tourists plan on their own, they face **too much online information** and many constraints, making it hard to choose the right services—especially in unfamiliar cities. If a system can quickly generate the **best travel route**, it will greatly improve the tourist experience and benefit the attractions as well.

The development of computer science and technology in recent years
has provided new ideas and methods to solve the travel route planning
problem (Zan, 2022). Many scholars have studied the travel route
problem through existing algorithms, including ant colony algorithm,
genetic algorithm, particle swarm algorithm and simulated annealing
algorithm (Mohammadzadeh & Gharehchopogh, 2020; Gharehchopogh
et al., 2023a; Gharehchopogh, 2023). The Ant colony

algorithm is a
probabilistic algorithm used to find the optimal path. It was inspired by
the way ants find their way around food. This algorithm has the char
acteristics of distributed computation, positive feedback of information
and heuristic search. It is a heuristic global optimization algorithm in
evolutionary algorithms (Gharehchopogh et al., 2023b).
This research fully absorbs the previous work achievements, and
proposes an intelligent tourism route planning model that combines Ant
colony optimization algorithms and genetic algorithm, and improves the
global search ability through cross operation and roulette wheel selec
tion. At the same time, smoothing and guiding factors are introduced to
improve the smoothness and operational efficiency of the model. The
main contribution of this research is to propose a new combination al
gorithm for path optimization, which provides ideas and methods for research on improving the
performance of Algorithmic efficiency, global
search capability, etc. At the same time, it also provides a feasible path
planning model for freelance tourists, which can quickly plan the
optimal travel path and time in complex environments, reduce tourists'
energy consumption, and improve their travel quality and experience.
The first part of the research describes the research status of path
planning and Ant colony optimization algorithms, and proposes
improvement methods according to the shortcomings of Ant colony
optimization algorithms in path planning. The second

part describes in
detail the path optimization process of the improved Ant colony opti
mization algorithms and the construction process of the multi-objective
tourism route planning model based on the algorithm. The third part
evaluates the efficiency and performance of the path planning of the
improved Ant colony optimization algorithms, and compares the
application effect of the tourism route planning model. The fourth part
summarizes the research process and experimental results, and proposes
improvement directions to address the shortcomings
.tell me it in short and precise

Here is a **short and precise** version:

Recent advances in computer technology have introduced better methods for solving travel route planning. Many algorithms—like ant colony, genetic, particle swarm, and simulated annealing—have been used. Ant colony optimization (ACO), inspired by ants finding food, is effective for finding optimal paths.

This study combines **ACO with genetic algorithms**, and adds **smoothing** and **guiding factors** to improve global search ability, speed, and path smoothness. The proposed model can quickly generate optimal travel routes in complex environments, reducing tourists' effort and improving their travel experience.

The paper is organized as follows:

1. Review of existing path planning and ACO limitations.
2. Explanation of the improved ACO and the multi-objective tourism planning model.
3. Performance evaluation and comparison with other algorithms.
4. Summary of findings and suggestions for future improvements.

With the growth of the Chinese economy, the development of the

Internet, and the improvement of travel conditions, it has led to an
increasing preference for free-wheeling forms of travel, such as self-
driving tours, RV tours, and backpacking (Zhang et al., 2020). Espe
cially after the epidemic, the economy has recovered and the tourism
industry has taken on a new vigor. Currently, there is even a rise in the
special forces-style and military training style of traveling to several sites
in one day, putting forward higher requirements for personalized travel
itinerary planning (Song et al., 2022). When tourists plan their travel
itineraries independently, they need to obtain relevant contents from a
large amount of network information. The problem of tourist informa
tion overload caused by the network sharing mechanism and the various
constraints that need to be considered in the itinerary make tourists
spend a lot of energy in screening the appropriate tourist services. This is
certainly a complex dilemma for tourists visiting unfamiliar cities. If an
optimal travel route can be quickly planned for a journey with many
attractions and a short duration, it will not only enhance the consumer's
journey experience, but also be an unbeatable benefit for the attractions.
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This research fully absorbs the previous work achievements, and
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main contribution of this research is to propose a new combination al
gorithm for path optimization, which provides ideas and methods for

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2667-3053/© 2023 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by nc-nd/4.0/).

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research on improving the performance of Algorithmic efficiency, global
search capability, etc. At the same time, it also provides a feasible path
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optimal travel path and time in complex environments, reduce tourists'
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application effect of the tourism route planning model. The fourth part
summarizes the research process and experimental results, and proposes
improvement directions to address the shortcomings.
tell me it i short and precie nda in summery

Here is a **short, precise, and clean summary** of the entire passage:

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### Short & Precise Summary

As China’s economy and internet use grow, more tourists prefer flexible travel styles like self-driving and backpacking. After COVID-19, tourism demand increased, and many people now try to visit several places in one day. This creates a need for **highly personalized and efficient travel route planning**.

However, tourists face **information overload** and many constraints, making it hard to plan routes—especially in unfamiliar cities. A system that can quickly generate the **best travel path** would greatly improve tourist experience and benefit attractions.

Advances in computer science have introduced new optimization methods. Many algorithms—such as ant colony, genetic, particle swarm, and simulated annealing—have been used. Ant Colony Optimization (ACO), inspired by how ants find food, is known for good global search ability.

This research proposes a new model that **combines ACO with Genetic Algorithm**, and adds **smoothing** and **guiding factors** to improve accuracy, speed, and global search performance. The model can quickly plan optimal tourist routes, save time, reduce effort, and improve overall travel experience.

The paper includes:

1. Review of existing path-planning and ACO limitations.
2. Description of the improved algorithm and multi-objective route planning model.
3. Performance evaluation and comparison with other algorithms.
4. Conclusion and future improvement directions.

---

If you want, I can also summarize the next section or make an ultra-short 2–3 line summary.

mproved ant colony algorithm for collision-free path planning
The planning of a personalized travel itinerary requires consider
ation of several factors such as tourist preferences, transportation
2

conditions and travel time. This process can be regarded as a combi
natorial optimization problem with multiple constraints. Ant colony
optimization (ACO) is usually chosen for solving such problems. ACO
reduces the travel route planning problem to a path finding process, i.e.,
finding the shortest collision-free path in a map with obstacles (Yang,
2022). The path optimization process of the ant colony algorithm is
shown in Fig. 1. First, grid maps are constructed, then the heuristic in
formation matrix is constructed, and the path, path length, heuristic
information matrix and tabu list are updated. The colony compares the
path with the original shortest path, and if the path is longer than the
shortest path, it continues to find the path. Instead, the ant colony finds
the shortest path and the algorithm stops iterating.
First, all parameters of the model are initialized. Then the obstacle
raster map with y rows and x columns is created, and the rasters in the
map are marked with the serial number (1,2,…,i)from left to right and
from bottom to top. If the side length of each grid is a, the coordinate
position of the i grid can be calculated by Eq. (1).
{xi=a∗[mod(i,y) a/2]
yi=a∗[x+a/2 ceil(i,x)] (1)
The mod and ceil represent the remainder function and the upward
rounding function, respectively. For the movement of ants in the raster
map, an eight-way steering strategy is used, i.e., ants can move from the i

raster to the adjacent eight rasters: top, bottom, left, right, top-right,
bottom-right, top-left, and bottom-left. Then based on the grid co
ordinates obtained from Eq. (1), the Di,j which is the distance between
the i-th grid and the adjacent grid j, can be calculated as shown in Eq.
(2).
Di,j=

xi xj
)2+ yi yj
)2
√
(2)
To make the ants move in a path avoiding obstacles, a heuristic in
formation matrix D is introduced to evaluate the feasibility of traveling
from each grid to the adjacent grid. The heuristic factor is assigned to 1
/Di,j for the unobstructed grid of i and 0 for the obstructed grid of j, and
the matrix D is entered. Then the ants can determine whether the eight
neighboring grids in D are obstacles or not based on the data in D. Since
the trajectory of ants in the ant colony algorithm is randomly distrib
uted, the probability of going from grid i to grid j in t iterations is
influenced by the pheromone ai,j(t)and the heuristic information. The
relevant formula is given in Eq. (3).tell me it in short and precise and easy keepign all necesary terms into it

Here is a **short, precise, and easy** summary while keeping all key technical terms:

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### Summary: Improved Ant Colony Algorithm for Collision-Free Path Planning

Planning a personalized travel route involves many constraints such as **tourist preferences, transportation conditions, and travel time**, making it a **combinatorial optimization problem**.
To solve this, the **Ant Colony Optimization (ACO)** algorithm is used, which turns travel planning into finding the **shortest collision-free path** on a map with obstacles.

The improved ACO works as follows:

1. **Create a grid (raster) map** with x columns and y rows.
   Each grid cell is numbered, and its coordinates $(x_i, y_i)$ are calculated using **Eq. (1)** with `mod` and `ceil` functions.
2. Ants move using an **8-direction strategy** (top, bottom, left, right, and all 4 diagonals).
3. The distance between two adjacent grids, $D_{i,j}$, is computed using the **Euclidean distance formula** (Eq. 2).
4. A **heuristic information matrix D** is created:
   - If grid j is free → heuristic = $1/D_{i,j}$
   - If grid j is an obstacle → heuristic = 0
5. Ants choose their next move based on:
   - **Pheromone level** $a_{i,j}(t)$
   - **Heuristic information**
     The transition probability is calculated using **Eq. (3)**.
6. During the search, the algorithm:
   - Updates the path and path length
   - Updates the heuristic matrix and tabu list
   - Compares the new path with the best (shortest) one
   - Stops when the shortest path is found

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If you want, I can also summarize the equations (1–3) in simple language.

tell me base papers  (Construction of intelligent planning model for tourism routes)this part with every necessary equation  and therir importance and their meaning

Below is a **clear, simple, and complete explanation** of the section **"Construction of Intelligent Planning Model for Tourism Routes"**, including **all necessary equations, their meaning, and importance**.

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# ✅ Construction of Intelligent Planning Model for Tourism Routes — Simplified Explanation

The goal is to build a **smart tourism route planning model** that finds the best travel path considering many real-world factors.

This section consists of two major parts:

1. **Improved Ant Colony Algorithm (ACO) for collision-free path planning**
2. **Tourism route planning model with multiple factors**

Below is each part explained **clearly with all required equations**.

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# ⭐ 1. Improved Ant Colony Algorithm (ACO)

Used to find the **shortest collision-free path** on a map with obstacles.

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## 🔹 Step 1: Create Grid Map + Coordinates

Each grid cell is assigned a number and a coordinate $(x_i, y_i)$.

### 📌 Equation (1): Grid coordinate calculation

$$\begin{cases} x_i = a \cdot [\mathrm{mod}(i, y) - a/2] \\ y_i = a \cdot [x + a/2 - \mathrm{ceil}(i, x)] \end{cases}$$

### ✔️ Meaning

- Converts grid number **i** into actual **(x, y)** position.
- `mod` → remainder, `ceil` → upward rounding.
- Important for calculating distances and obstacles.

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## 🔹 Step 2: Calculate Distance Between Grids

📌 **Equation (2): Euclidean distance**

$$D_{i,j} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$$

✔️ **Meaning & Importance**

- Gives distance between grid i and grid j.
- Needed to encourage ants to move toward **shorter paths**.

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## 🔹 Step 3: Build Heuristic Information Matrix (D)

- If grid j is **not an obstacle**, heuristic = $1/D_{i,j}$
- If grid j is **an obstacle**, heuristic = 0

✔️ **Importance**

- Helps ants avoid obstacles.
- Guides them to shorter steps.

---

## 🔹 Step 4: Probability of Ant Moving to Next Grid

📌 **Equation (3): Transition probability**

$$p_{i,j}^k(t) = \frac{[a_{i,j}(t)]^\alpha \cdot [1/D_{i,j}]^\beta}{\sum_{j \in J_k(i)} [a_{i,j}(t)]^\alpha \cdot [1/D_{i,j}]^\beta}$$

✔️ **Meaning**

Probability that **ant k** moves from grid i → j.

✔️ **Importance**

- Influenced by **pheromone strength** & **distance**.
- Directly affects path exploration.

---

## 🔹 Step 5: Pheromone Update

📌 **Equation (4): Pheromone evaporation + update**

$$a_{i,j}(t+n) = (1-\rho)a_{i,j}(t) + \sum_k \Delta a_{i,j}^k$$

### ✔️ Meaning

- Pheromone **evaporates** over time (controlled by ρ).
- New pheromone is added from ants' path.

### ✔️ Why Important

Prevents ants from getting stuck in bad paths.

---

## 🔹 Step 6: Pheromone Increase for Each Ant

### 📌 Equation (5): Pheromone increment

$$\Delta a_{i,j}^{k} = \begin{cases} Q/L_k, & \text{if ant k passed i} \rightarrow \text{j} \\ 0, & \text{otherwise} \end{cases}$$

### ✔️ Meaning

- More pheromone is placed on **shorter paths** (because Lk is small).
- Longer paths get less pheromone.

### ✔️ Importance

Encourages global optimization.

---

## 🔹 Step 7: Improve Probability Formula with Smoothing & Guiding Factors

### 📌 Equation (6): Improved transition probability

$$p_{i,j}^{k}(t) = \frac{[a_{i,j}(t)]^{\alpha}[1/D_{i,j}]^{\beta}[\lambda_{i,j}(t)]^{\phi}[\omega_{i,j}(t)]^{\gamma}}{\sum_{j \in J_k(i)}[\ldots same \ldots]}$$

### ✔️ Meaning

Adds two new factors:

1. **ω = smoothing factor** → reduces sharp turns
2. **λ = guiding factor** → directs ants faster towards goal

### ✔️ Importance

- Faster convergence
- Smoother and more natural paths

- Better global search

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### 🔹 Step 8: Smoothing Factor

#### 📌 Equation (7): Angle-based smoothing

$$\omega_{i,j}(t) = \begin{cases} 1, & \theta = 0^\circ \\ 2, & \theta = 45^\circ \\ 3, & \theta = 90^\circ \\ 4, & \theta = 135^\circ \end{cases}$$

#### ✔️ Meaning

Larger angle → more cost

Encourages straighter paths.

---

### 🔹 Step 9: Guiding Factor

#### 📌 Equation (8): Bootstrap (guiding) factor

$$\lambda_{i,j}(t) = \frac{k}{K} \cdot \frac{t}{t_{max}} \cdot \frac{1}{D_{i,E} + D_{i,j}}$$

#### ✔️ Meaning

- Uses iteration number t
- Uses distance to end point $D_{i,E}$
- Grows stronger over time → “pushes” ants toward goal

#### ✔️ Importance

- Greatly improves efficiency
- Reduces search randomness

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## ⭐ 2. Multi-Objective Tourism Route Planning Model

After basic path planning, the model considers **real tourist factors**:

✔️ Travel time

✔️ Crowding

✔️ Tourist preference
✔️ Satisfaction
✔️ Total tour time

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## 🔹 Travel Time Factor

### 📌 Equation (9): Travel time

$$M_{i,j} = \frac{D_{i,j}}{V_{i,j}}$$

### ✔️ Meaning

Travel time = distance / travel speed.

### ✔️ Importance

Slower routes get lower priority.

---

## 🔹 Crowding Factor

### 📌 Equation (10): Crowding level

$$C_j = \frac{E_j}{A_j}$$

### ✔️ Meaning

- $E_j$ = number of visitors
- $A_j$ = max capacity
  Higher value = more crowded.

---

## 🔹 Tourist Preference Factor

### 📌 Equation (11): Preference

$$H_j = s_j \cdot g_j$$

### ✔️ Meaning

- $s_j$ = similarity between tourist and attraction

- $g_j$ = average online rating
  Higher Hj = more preferred attraction.

---

## 🔹 Combined Travel Probability

### 📌 Equation (12): Multi-factor transition probability

$$P_{i,j} = \frac{(1/M_{i,j})^{\varepsilon} \cdot (1/C_j)^{\tau} \cdot H_j^{\delta}}{\sum_{j \in J_k(i)} (1/M_{i,j})^{\varepsilon} \cdot (1/C_j)^{\tau} \cdot H_j^{\delta}}$$

### ✔️ Meaning

Considers 3 key factors:

- travel time
- crowding
- preference

---

## 🔹 Tourist Satisfaction

### 📌 Equation (13): Satisfaction

$$r_j = \frac{H_j}{M_{i,j} \cdot C_j}$$

### ✔️ Meaning

High preference + low time + low crowding = high satisfaction.

---

## 🔹 Overall Route Satisfaction

### 📌 Equation (14): Total satisfaction R

$$R = \max \sum_{i=1}^{n-1} \sum_{j=1}^{n-1} z_i r_i$$

### ✔️ Meaning

Finds route with **maximum total satisfaction**.

---

## 🔹 Constraints

### 📌 Equation (15): No repeating visits

Ensures each attraction is visited **at most once**.

---

## 🔹 Tour Time Allocation

### 📌 Equation (16): Time at attraction j

$$T_j = T \cdot \frac{H_j^{3/2}}{\sum H_j^{3/2}}$$

### ✔️ Meaning

More preferred attractions get more time.

---

# 🎉 Final Summary

This section creates an intelligent tourism route planning model by:

### ✔️ Improving ACO using

- smoothing factor
- guiding factor
- genetic algorithm

### ✔️ Adding real-world tourism factors

- travel time
- crowding
- preference
- satisfaction
- tour duration

This allows the algorithm to produce **fast, smooth, realistic, and tourist-friendly travel routes**.

---

If you want, I can also make:

✅ A very short exam-friendly summary
✅ A flowchart explanation
✅ Simplified meanings of all equations in one table

what is cellular structure?

A **cellular structure** simply means **dividing an area into small, equal-sized cells (grids)** — just like a chessboard or pixel map.

### ✔️ In the context of path planning / ant colony algorithm

A *cellular structure* is a way of representing a map by splitting it into small square cells. Each cell has information such as:

- Is it **free** or **an obstacle**?
- What are its **coordinates**?
- What **neighbors** (top, bottom, left, right, diagonals) it has?

### ✔️ Why cellular structure is used?

Because it makes path planning easier:

- You convert the map into small grid units.
- Ants (or robots) move **cell-by-cell**.
- The algorithm checks obstacles easily.
- Distance and movement directions can be calculated precisely.

### ✔️ Simple example

Imagine a city map divided into a 10×10 grid:

```
□ □ ■ □ □
□ □ □ ■ □
■ □ □ □ □
□ ■ □ □ □
```

- □ = free cell
- ■ = obstacle

The ant moves from one cell to the next, avoiding obstacle cells.

### ✔️ In short:

**Cellular structure = map divided into small cells used for calculating movement, distance, and obstacles.**

If you want, I can also explain how it helps ACO or draw a simple diagram.

The initial pheromone was obtained by individuals of the genetic algo
rithm during the initialization of the pheromone. Then the routes taken
by the ants are recorded using cellular structures instead of taboo tables.
By crossover operation of the genetic algorithm, some of the routes
taken by the ants are crossed over to obtain new individuals. This
method can expand the population sample with less number of itera
tions. Finally, the roulette wheel selection method of the genetic algo
rithm is introduced to find the probability of an individual being
selected according to the ratio of individual fitness to the sum of fitness
of all individuals in the population. And the individuals with probability
greater than the set value are used as new individuals to form a new
population. The process of ant colony algorithm (Genetic
algorithm-Path smoothing ant colony optimization, GA-PSACO) that
changes the probability and incorporates genetic algorithm is shown in
Fig. 2. The algorithm first builds a grid map, finds all accessible grids for
the next step, and then records each ant's path with a cellular structure.
The algorithm combines selection and crossover

operations with genetic
algorithms to calculate the path length, update the ant state and the
pheromone matrix until the target node is reached. The path is then
compared with the original shortest path, and if it is shorter than the
original path length, the shortest path length is updated. Otherwise,
iteration continues until the termination condition is met and the
shortest path and path length are output.  i think that this is the overall proess  and in it where the smoothing factor and bootstrap factr and corner are used tell me it in simple and easy adn in short and in precise

Here is a **simple, short, and precise** explanation of the **overall GA-PSACO process**, including **where smoothing factor, bootstrap (guiding) factor, and corner/angle factor are used**:

---

## ⭐ Overall GA-PSACO Process (Simple & Clear)

1. **Build grid map** (cellular structure).
2. **Initialize pheromone** using **Genetic Algorithm (GA)**.
3. Each ant starts moving on the grid → 8 possible directions.
4. When choosing the next grid, the ant uses **improved probability** that includes:
   - pheromone
   - distance
   - **smoothing factor** (discourages sharp turns)
   - **bootstrap / guiding factor** (pushes ants toward target)
   - corner/angle factor (similar to smoothing)
5. Ants record their path using **cellular structures** (not tabu list).
6. GA operations are used:
   - **crossover** → creates new better paths
   - **roulette wheel selection** → keeps high-fitness (short) paths
7. Update pheromone matrix.
8. Compare path with current shortest path.
9. If shorter → update shortest path.

10. Repeat until termination.

---

## ⭐ Where smoothing, guiding (bootstrap), and corner factors are used?

### ✅ They are used in the probability formula when an ant chooses its next step.

This formula determines **which grid the ant moves to**.

### ✔️ The smoothing factor (ω)

- Penalizes sharp turns.
- Encourages smoother, more natural paths.

### ✔️ The corner/angle factor

- Same idea:
- Avoids unnecessary zig-zag movement.

### ✔️ The bootstrap / guiding factor (λ)

- Pushes ants toward the goal.
- Makes ants converge faster instead of wandering randomly.

---

## ⭐ Simple Summary

- **Grid map** → where ants move
- **GA initializes pheromone & creates better paths**
- **Smoothing & corner factors** → keep path smooth
- **Guiding factor** → directs ants to goal quickly
- **GA crossover & selection** → improves population
- **Ants update pheromone** → best paths reinforced
- **Stop when shortest path is found**

---

If you want, I can also draw a very easy flowchart of GA-PSACO.